MCP MANUAL EN

Printix MCP — User Handbook

> **Version:** 6.8.10 · **Tool inventory:** 127 tools · **As of:** April 2026 **Audience:** Administrators, helpdesk and power users who interact with the Printix MCP Server through an AI assistant (claude.ai, ChatGPT, Claude Desktop, Cursor, etc.). **Language:** English · German version see `MCP_MANUAL_DE.pdf`

⚠️ Important: keep your AI assistant's tool list up to date

When the MCP server gets a new release, **new tools** are usually added. To make sure your AI assistant actually picks them up, the tool list has to be reloaded. **A server upgrade alone is not enough** — the client caches tool definitions.

Client	How to refresh the tool list
claude.ai (web)	*Settings → Connectors → Printix MCP → disconnect → reconnect*. Or just start a **new conversation** — the tool list is pulled on the first message.
ChatGPT (custom connector)	In the *Custom GPT editor*, click *Disconnect* on the MCP server, then *Connect*. Closing and re-opening the tab usually works too.
Claude Desktop	**Full app restart** (`Cmd+Q` then re-launch — *not* just close the window). Tools are loaded on startup.
Cursor / Continue / others	Toggle the connector off and on, or use the client's `/mcp reload` (varies by client).

Quick test to see if a new tool has arrived: ask your assistant "*Which Printix tools do you have?*". If you see e.g. `printix_print_self` or `printix_welcome_user` in the list, you're current. If not, do the refresh above.

What is Printix MCP?

The Printix MCP Server bridges modern AI assistants and the Printix Cloud Print API. It exposes **127 tools** that let you drive Printix in natural language — from the simple "*Which printers do we have in Düsseldorf?*" to complex workflows like "*Send me this weekly report as a PDF to my printer and archive a copy in Paperless tagged 'Q1-report'.*"

You **don't have to memorize tool names**. The assistant picks the right tool for your question. This handbook shows you *what's possible* so you can ask focused questions.

How to read this handbook

Each category has a short intro, a tool table with purpose descriptions and several **example dialogues** with concrete prompts and tool calls. You can use the prompts verbatim or as inspiration.

🆕 marks tools added in v6.8.0–v6.8.10 (April 2026).

Table of Contents

1. System & Self-Diagnosis
2. Printers, Sites & Networks
3. Print Jobs & Cloud Print
4. Users, Groups & Workstations
5. Cards & Card Profiles
6. Reports & Analytics
7. Report Templates & Scheduling
8. Capture & Document Workflow
9. Onboarding, Time-Bombs & Entra Sync 🆕
10. Operations, Maintenance & Audit
11. Tips for productive AI dialogues

---

1. System & Self-Diagnosis

Meta questions: *Who am I? Is everything running? What's my role? What should I do next?* Great as a session opener — or when something doesn't work and you need to know **why**.

Tool	Purpose
`printix_status`	Health check: server up, tenant reachable, configured credential scopes.
`printix_whoami`	Current tenant + own Printix user + admin status.
`printix_tenant_summary`	Snapshot: printers, users, sites, cards, open jobs counts.
`printix_explain_error`	Translates a Printix error code or message to plain English + remedy hints.
`printix_suggest_next_action`	Suggests a sensible next step based on a context string.
`printix_natural_query`	Takes a natural-language question and proposes the matching reports tool.

Example dialogues

Prompt: "*Is Printix up?*" → `printix_status` reports API connection, tenant ID and configured credential scopes.

Prompt: "*Who am I logged in as in Printix?*" → `printix_whoami` returns tenant, your email and admin flag.

Prompt: "*Give me an overview of my tenant.*" → `printix_tenant_summary` returns all key counts in one block.

Prompt: "*What does error 'AADSTS700025' mean?*" → `printix_explain_error("AADSTS700025")` explains it (public client / no client_secret with PKCE) and suggests fixes.

Prompt: "*What should I do next? I just installed a new printer.*" → `printix_suggest_next_action("new printer installed")` suggests SNMP config check, test print, health report.

---

2. Printers, Sites & Networks

Physical and logical infrastructure: printers, queues, sites, networks, SNMP configs. Read and write operations. The `*_context` tools return aggregated views (queue + printer + recent jobs in one call).

Tool	Purpose
`printix_list_printers`	All printers (optional search).
`printix_get_printer`	Details + capabilities of a specific printer.
`printix_resolve_printer`	Fuzzy match (name + location + model + site).
`printix_network_printers`	All printers in a network or site.
`printix_get_queue_context`	Queue + printer object + recent jobs in one call.
`printix_printer_health_report`	Status grouped: online / offline / error states.
`printix_top_printers`	Top-N printers by volume.
`printix_list_sites` / `printix_get_site`	Site listing / details.
`printix_create_site` / `printix_update_site` / `printix_delete_site`	Site management.
`printix_site_summary`	Site + networks + printers in one aggregated block.
`printix_list_networks` / `printix_get_network`	Network listing / details.
`printix_create_network` / `printix_update_network` / `printix_delete_network`	Network management.
`printix_get_network_context`	Network + site + printers in one block.
`printix_list_snmp_configs` / `printix_get_snmp_config`	SNMP configs.
`printix_create_snmp_config` / `printix_delete_snmp_config`	SNMP config create/delete.
`printix_get_snmp_context`	SNMP config + affected printers + network.

Example dialogues

Prompt: "*Which Brother printers are in Düsseldorf?*" → `printix_resolve_printer("Brother Düsseldorf")` token-fuzzy match across name/location/vendor/site.

Prompt: "*Show me all printers in network 9cfa4bf0.*" → `printix_network_printers(network_id="9cfa4bf0")` resolves the site (when no direct network→printer mapping exists) and returns the relevant printers.

Prompt: "*Give me a complete summary of the DACH site.*" → `printix_site_summary(site_id=…)` — site meta + networks + all printers.

Prompt: "*Which printers are currently offline?*" → `printix_printer_health_report` groups by status, problems first.

Prompt: "*Top 5 printers by page count last week?*" → `printix_top_printers(days=7, limit=5, metric="pages")`.

Prompt: "*Create a new site 'Hamburg' at Mönckebergstraße 7.*" → `printix_create_site(name="Hamburg", address="Mönckebergstraße 7", ...)`.

---

3. Print Jobs & Cloud Print

View, submit and delegate print jobs. 🆕 **v6.8.x**: three new high-level tools (`print_self`, `print_to_recipients`, `session_print`) for AI workflows with native PDF/PCL conversion.

> 🆕 **Auto PDL conversion (v6.8.8+)**: All print tools auto-convert PDF/PostScript/Text to PCL XL via Ghostscript before submitting to the printer queue. Without this, printers without a PDF RIP would print hieroglyphs (raw PDF source as ASCII). Default param `pdl="auto"` (= PCL XL color). Use `pdl="passthrough"` to send the file unchanged.

Tool	Purpose
`printix_list_jobs`	All jobs, optional queue filter.
`printix_get_job`	Job details.
`printix_submit_job`	Low-level job submit (stage 1 of the 5-stage flow).
`printix_complete_upload`	Finish upload.
`printix_delete_job`	Cancel job.
`printix_change_job_owner`	Delegate job to another user.
`printix_jobs_stuck`	Jobs hanging more than N minutes.
`printix_quick_print`	Single-shot: URL + recipient → done.
`printix_send_to_user`	Send document (URL or base64) to user X. v6.8.8+: with auto-conversion.
🆕 `printix_print_self`	Print to **own** secure-print queue (AI generates PDF inline).
🆕 `printix_print_to_recipients`	Multi-recipient: one PDF to many recipients (also via `group:Name` or `entra:OID`).
🆕 `printix_resolve_recipients`	Diagnostic: shows which recipients would resolve before `print_to_recipients`.
🆕 `printix_session_print`	Job with time-bomb (auto-expire after N hours).

Example dialogues

Prompt: "*Send this PDF as Secure Print to marcus@firma.de.*" → `printix_send_to_user(user_email="marcus@firma.de", file_content_b64=..., filename="contract.pdf")` — server converts PDF→PCL XL and queues the job.

Prompt: "*Which jobs have been stuck for more than 30 minutes?*" → `printix_jobs_stuck(minutes=30)`.

Prompt: "*Hand over job 4711 to marcus@firma.de — I'm going on vacation.*" → `printix_change_job_owner(job_id="4711", new_owner_email="marcus@firma.de")`.

🆕 **Prompt:** "*Build me an A4 print template with the quarterly numbers and queue it for me to pick up at the printer.*" → AI generates PDF inline → `printix_print_self(file_b64=..., filename="Q1.pdf")` puts the job in your own queue. Pick it up at the printer with card/code.

🆕 **Prompt:** "*Send this memo to everyone in the Marketing group as Secure Print.*" → `printix_print_to_recipients(recipients_csv="group:Marketing", file_b64=..., filename="memo.pdf")` — one job per member.

🆕 **Prompt:** "*Before I send: who exactly is on the list?*" → `printix_resolve_recipients("group:Marketing, alice@firma.de, entra:abc-uuid")` shows the resolved recipients without printing. Ambiguities + not-found inputs are listed separately.

🆕 **Prompt:** "*Send this draft contract to externer-gast@partner.de — should auto-expire after 4 hours.*" → `printix_session_print(user_email="externer-gast@partner.de", file_b64=..., filename="contract.pdf", expires_in_hours=4)` — submits the job and arms a cleanup time-bomb.

🆕 **Prompt:** "*Print mono only — no need for color.*" → Set `color=False` on any print tool (default `True`).

🆕 **Prompt:** "*My printer does the PDF RIP itself — just pass the file through.*" → Set `pdl="passthrough"` to disable server-side conversion.

---

4. Users, Groups & Workstations

Full lifecycle: create, edit, deactivate, diagnose. The `user_360` and `diagnose_user` tools are helpdesk all-rounders. 🆕 **v6.8.x**: detailed group membership + print pattern analysis + quota guard.

Tool	Purpose
`printix_list_users`	All users, with pagination + role filter.
`printix_get_user`	User details.
`printix_find_user`	Fuzzy search by email or name.
`printix_user_360`	360° view: user + cards + groups + workstations + recent jobs.
`printix_diagnose_user`	Helpdesk diagnosis: what works, what doesn't, why.
`printix_create_user` / `printix_delete_user`	User management.
`printix_generate_id_code`	New ID code for a user.
`printix_onboard_user` / `printix_offboard_user`	Guided on/offboarding (multi-step in one call).
`printix_list_admins`	All admins.
`printix_permission_matrix`	Matrix: user × permissions.
`printix_inactive_users`	Users idle for N days.
`printix_sso_status`	Check SSO mapping.
`printix_list_groups` / `printix_get_group`	Group listing / details.
`printix_create_group` / `printix_delete_group`	Group management.
🆕 `printix_get_group_members`	Members of a group by UUID or name (ambiguity-safe).
🆕 `printix_get_user_groups`	Reverse lookup: which groups is user X in?
🆕 `printix_describe_user_print_pattern`	Print profile of a user: top printers, color quote, page count.
🆕 `printix_quota_guard`	Pre-flight burst check before submit (verdict allow/throttle/block).
🆕 `printix_print_history_natural`	History with natural-language windows ("today", "last week", "Q1", "7d").
`printix_list_workstations` / `printix_get_workstation`	Workstations listing / details.

Example dialogues

Prompt: "*Tell me everything you know about marcus@firma.de.*" → `printix_user_360(query="marcus@firma.de")` returns the full 360° view.

Prompt: "*Why can't Anna print anymore?*" → `printix_diagnose_user(email="anna@firma.de")` checks status, SSO, cards, groups, blockers — and suggests fixes.

Prompt: "*Which users have been inactive for 180 days?*" → `printix_inactive_users(days=180)` — offboarding candidate list.

Prompt: "*Onboard a new employee: peter@firma.de, Peter Meier, group 'Finance'.*" → `printix_onboard_user(...)` runs all steps in order.

🆕 **Prompt:** "*Who's in the Marketing group?*" → `printix_get_group_members("Marketing")` — for ambiguous names you get a candidate list with UUIDs.

🆕 **Prompt:** "*Which groups is Anna in?*" → `printix_get_user_groups("anna@firma.de")` — checks user object first, falls back to group scan.

🆕 **Prompt:** "*What did Marcus print today?*" → `printix_print_history_natural(user_email="marcus@firma.de", when="today")`.

🆕 **Prompt:** "*Marcus's print pattern over the last 30 days?*" → `printix_describe_user_print_pattern(user_email="marcus@firma.de", days=30)` — top printers, color quote, average pages.

🆕 **Prompt:** "*Before I send another PDF: has the user not sent too many in the last 5 minutes?*" → `printix_quota_guard(user_email="marcus@firma.de", window_minutes=5, max_jobs=10)` — verdict `allow` / `throttle` / `block`.

---

5. Cards & Card Profiles

Everything around RFID/Mifare/HID cards: registration, mapping, profiles, bulk import. 🆕 **v6.8.x**: AI wrapper `card_enrol_assist` that does UID + profile-transform + register in one tool.

Tool	Purpose
`printix_list_cards`	Cards of a user.
`printix_list_cards_by_tenant`	All cards in tenant (filter: `all` / `registered` / `orphaned`).
`printix_search_card`	Search card by ID/number.
`printix_register_card`	Assign card to user (low-level).
`printix_delete_card`	Remove card assignment.
`printix_get_card_details`	Card + local mapping + owner in one block.
`printix_decode_card_value`	Decode raw card value (Base64/Hex/YSoft/Konica variants).
`printix_transform_card_value`	Run value through transformation pipeline (hex↔dec, reverse, prefix/suffix …).
`printix_get_user_card_context`	User + all cards + profiles in one block.
`printix_list_card_profiles` / `printix_get_card_profile`	Profile listing/details.
`printix_search_card_mappings`	Search local mapping DB.
`printix_bulk_import_cards`	CSV bulk import (with profile + dry run).
`printix_suggest_profile`	Suggest profile from sample UID (top-10).
`printix_card_audit`	Audit trail of all card changes for a user.
`printix_find_orphaned_mappings`	Local mappings without matching Printix user.
🆕 `printix_card_enrol_assist`	AI onboarding: UID + profile transform + register in one call.

Example dialogues

Prompt: "*Which cards does Marcus have?*" → `printix_get_user_card_context(email="marcus@firma.de")` — user + all cards + profiles.

Prompt: "*What is card UID* `04:5F:F0:02:AB:3C` *?*" → `printix_decode_card_value(card_value="04:5F:F0:02:AB:3C")` recognizes hex with separators, returns decoded bytes + profile hint.

Prompt: "*Import 500 cards from this CSV — but first as a dry run.*" → `printix_bulk_import_cards(csv=..., profile=..., dry_run=True)` validates each row and shows previews without writing.

Prompt: "*For UID* `045FF002` *— which profile fits?*" → `printix_suggest_profile(sample_uid="045FF002")` — top-10 with score + `best_match`.

🆕 **Prompt:** "*Marcus tapped his ID card on the iPhone. UID is* `04A1B2C3D4E5F6`*. Please register it for him.*" → `printix_card_enrol_assist(user_email="marcus@firma.de", card_uid_raw="04A1B2C3D4E5F6")` — runs through the default card profile and registers it via `register_card`.

---

6. Reports & Analytics

Reports run against the separate SQL Server warehouse. You get key figures, trends, anomalies and ad-hoc queries through a unified interface. `query_any` is the universal entry point; the specialized tools are quicker shortcuts for common questions.

Tool	Purpose
`printix_reporting_status`	Reports engine status (DB connection, last nightlies, preset count).
`printix_query_any`	Universal: preset + filters → table.
`printix_query_print_stats`	Print volume by dimension.
`printix_query_cost_report`	Costs, optionally by department/user.
`printix_query_top_users` / `printix_query_top_printers`	Top-N with time window.
`printix_query_anomalies`	Anomaly detection.
`printix_query_trend`	Trend lines over time.
`printix_query_audit_log`	Structured audit trail of the MCP server (actions, objects, actor).
`printix_top_printers` / `printix_top_users`	Short-form wrappers.
`printix_print_trends`	Trend by day/week/month.
`printix_cost_by_department`	Costs aggregated per department.
`printix_compare_periods`	Period A vs period B.

Example dialogues

Prompt: "*Who printed the most last month?*" → `printix_top_users(days=30, limit=10, metric="pages")`.

Prompt: "*Show the print trend for the last 90 days, monthly.*" → `printix_print_trends(group_by="month", days=90)`.

Prompt: "*Compare the last 30 days with the 30 days before — what changed?*" → `printix_compare_periods(days_a=30, days_b=30)` returns delta KPIs.

Prompt: "*Which department has the highest print costs?*" → `printix_cost_by_department(days=30)`.

Prompt: "*What did user X do in the MCP on April 15?*" → `printix_query_audit_log(start_date="2026-04-15", end_date="2026-04-15", actor_email="x@firma.de")`.

Prompt: "*Are there any anomalies in print behavior the last 14 days?*" → `printix_query_anomalies(days=14)` — e.g. sudden volume spikes or unusual printer usage patterns.

---

7. Report Templates & Scheduling

For analyses you need regularly: save as template, schedule recurring delivery, ship by email. Design options via `list_design_options`; `preview_report` renders a PDF preview without sending.

Tool	Purpose
`printix_save_report_template`	Save query + design as template.
`printix_list_report_templates`	All saved templates.
`printix_get_report_template`	Template details.
`printix_delete_report_template`	Delete template.
`printix_run_report_now`	Run template once, deliver.
`printix_send_test_email`	Test email (SMTP check).
`printix_schedule_report`	Schedule template as cron job.
`printix_list_schedules`	All active schedules.
`printix_update_schedule` / `printix_delete_schedule`	Schedule modify/delete.
`printix_list_design_options`	Available color schemes, logos, layouts.
`printix_preview_report`	PDF preview of a report without sending.

Example dialogues

Prompt: "*Save the current top-10 user report as template 'Monthly Print Top10'.*" → `printix_save_report_template(...)`.

Prompt: "*Send this template on the 1st working day of every month to management@firma.de.*" → `printix_schedule_report(report_id=…, cron="0 8 1 * *", recipients=["management@firma.de"])`.

Prompt: "*Show me a PDF preview of template XY.*" → `printix_preview_report(report_id=…)`.

Prompt: "*Which color schemes can I use for reports?*" → `printix_list_design_options()`.

---

8. Capture & Document Workflow

Capture connects scanned or AI-generated documents to target systems (Paperless-ngx, SharePoint, DMS …) via plugins. 🆕 **v6.8.x**: two new tools that let the AI feed files directly into a capture workflow — not only via the classic webhook path.

Tool	Purpose
`printix_list_capture_profiles`	All capture profiles in tenant.
`printix_capture_status`	Status: server port, webhook base URL, available plugins, configured profiles.
🆕 `printix_describe_capture_profile`	Plugin schema of a profile: which metadata fields are accepted, plus current config (secrets masked).
🆕 `printix_send_to_capture`	Feed file directly into capture workflow — same code path as a webhook, no printer/blob detour.

Example dialogues

Prompt: "*Is capture active and which plugins are installed?*" → `printix_capture_status` — plugin list (e.g. paperless_ngx) + count of configured profiles.

Prompt: "*Which capture profiles do I have?*" → `printix_list_capture_profiles` — list with target system, webhook URL and recent runs.

🆕 **Prompt:** "*What metadata does the Paperless profile accept?*" → `printix_describe_capture_profile("Paperless (Marcus)")` — plugin schema (`tags`, `correspondent`, `document_type` etc.) plus current default values.

🆕 **Prompt:** "*Save this AI-generated contract to Paperless with tags 'Q1', 'Contract' and correspondent 'Acme Corp'.*" → `printix_send_to_capture(profile="Paperless (Marcus)", file_b64=..., filename="contract_acme.pdf", metadata_json='{"tags":["Q1","Contract"], "correspondent":"Acme Corp", "document_type":"Contract"}')`.

🆕 **Prompt:** "*My weekly report is done — file it in Paperless and send a print copy to my printer too.*" → Two calls: `printix_send_to_capture(...)` + `printix_print_self(...)` — the assistant chains both automatically.

---

9. Onboarding, Time-Bombs & Entra Sync 🆕

Everything in this section is **new in v6.8.x**. It's about automated user lifecycle workflows: a freshly created user receives a welcome PDF, reminder time-bombs are scheduled, and 7 days later we follow up automatically if no action has been taken. Plus: Entra/AD group sync with Printix groups.

Tool	Purpose
🆕 `printix_welcome_user`	Onboarding companion: welcome PDF + time-bombs for `card_enrol`, `first_print_reminder`.
🆕 `printix_list_timebombs`	View active/past time-bombs in tenant.
🆕 `printix_defuse_timebomb`	Manually defuse a time-bomb (with audit reason).
🆕 `printix_sync_entra_group_to_printix`	Diff Entra/AD group members vs Printix group members (default `report_only`).

Example dialogues

🆕 **Prompt:** "*Onboard new employee peter@firma.de and run the welcome workflow with reminders.*" → `printix_onboard_user(...)` (for DB/account creation) + `printix_welcome_user(user_email="peter@firma.de")` puts a welcome PDF in his secure-print queue AND arms two time-bombs: 3-day reminder if no first print, 7-day reminder if no card enrolled.

🆕 **Prompt:** "*Which onboarding reminders are currently active?*" → `printix_list_timebombs(status="pending")`.

🆕 **Prompt:** "*Peter just told me he's on holiday until next week. Defuse his reminders.*" → `printix_list_timebombs(user_email="peter@firma.de")` shows his bombs with IDs, then `printix_defuse_timebomb(bomb_id=42, reason="on holiday until 2026-05-12")`.

🆕 **Prompt:** "*Sync Entra group `Marketing-DACH` (OID `abc-123`) with our Printix group — but report only, no writes.*" → `printix_sync_entra_group_to_printix(entra_group_oid="abc-123", printix_group_id="def-456", sync_mode="report_only")` returns diff: `to_add` (in Entra but not Printix) and `to_remove` (in Printix but not Entra).

> ℹ️ **Prerequisite for `sync_entra_group_to_printix`**: the MCP server's Entra app registration needs the application permission `Group.Read.All` (Application, not Delegated) + admin consent. Add it in the Azure portal.

Time-bomb concept

Time-bombs are **conditional, deferred actions**. A cron job runs hourly (`minute=7`) and for every pending bomb checks:

1. Is the condition still met? (e.g. "*user has no card enrolled yet*")
2. If yes → run the action (queue a reminder PDF, log entry, etc.).
3. If no → bomb is auto-marked `defused`.

So the `first_print_reminder_3d` bomb stays pending after 3 days — if the user actually printed in between, it auto-defuses. Only if the user really did nothing, the reminder fires.

---

10. Operations, Maintenance & Audit

Backups, demo data, feature tracking. Mix of operations and meta.

Tool	Purpose
`printix_list_backups`	All existing backups.
`printix_create_backup`	New backup (DB + config + metadata).
`printix_demo_setup_schema`	Create demo schema in reports DB.
`printix_demo_generate`	Generate synthetic demo data.
`printix_demo_rollback`	Remove demo data (by demo tag).
`printix_demo_status`	Which demo sets are active?
`printix_list_feature_requests` / `printix_get_feature_request`	Ticket system for feature requests.

Example dialogues

Prompt: "*Make a backup before I change anything.*" → `printix_create_backup`.

Prompt: "*Set up a demo environment with 50 users and 500 jobs.*" → `printix_demo_setup_schema` (once) + `printix_demo_generate(users=50, jobs=500)`.

Prompt: "*Show me all open feature requests.*" → `printix_list_feature_requests(status="open")`.

---

11. Tips for productive AI dialogues

1. **Think in goals, not tools.** "*Who prints too much?*" beats "*call query_top_users with days=30*". The assistant picks the right tool.
2. **Provide context.** "*Marcus from finance*" is clearer than just "*Marcus*".
3. **Use the 360° tools.** `printix_user_360`, `printix_get_queue_context`, `printix_site_summary` save you follow-up questions.
4. **Ask "why" on errors.** "*Why did this fail?*" triggers `printix_explain_error` or `printix_diagnose_user`.
5. **Dry run before bulk operations.** `printix_bulk_import_cards(dry_run=True)`, `printix_resolve_recipients` (before `print_to_recipients`), `printix_sync_entra_group_to_printix(sync_mode="report_only")`.
6. 🆕 **Multi-step workflows in a single prompt.** The assistant can chain tool calls: "*Generate a weekly report PDF, archive it in Paperless, and print a copy for me.*"
7. 🆕 **Trust auto-conversion in `print_*` tools.** Default `pdl="auto"` does the right thing 99% of the time (PCL XL color). Only override when your printer queue does conversion server-side (`passthrough`) or you explicitly need PostScript.
8. 🆕 **Refresh the tool list regularly** (see notice at the top). Otherwise the assistant uses outdated tool definitions.

Document generated from Printix MCP Server v6.8.10 · April 2026 · 127 tools · Repository